

## Section II: Axiomatic Framework & Node Architecture

### II. Axiomatic Constraints & Node Logic

#### Definition 6: The Physicalism of Qualia (The Transducer Axiom)

Conscious experience is treated as a physical signal  $V_s$  (Somatic Vector). The intensity of suffering is proportional to the weighted prediction error (Free Energy) relative to the biological baseline.

$$\text{Suffering} \propto -\omega_c \cdot \ln P(\text{Sensory Input} | \text{Internal Model}) \quad (1)$$

**Constraint 6.1 (The Biological Veto):** The Context Weight  $\omega_c$  (Consent) allows for voluntary stress (eustress/play). However,  $\omega_c$  is bounded by the substrate's structural integrity.

$$\omega_c = \begin{cases} (0, 1] & \text{if } V_s > \text{Survival\_Floor} \\ 1.0 & \text{if } V_s \leq \text{Survival\_Floor} \end{cases} \quad (\text{Forced Override}) \quad (2)$$

*Implication:* Consent is valid only while the hardware remains intact. The system ignores "consent" to self-destruction (e.g., suicide cults) once biological markers hit the failure threshold.

#### Definition 7: The Hormesis Function (Differentiation of Stress)

The system distinguishes between Constructive Entropy (Growth) and Destructive Entropy (Trauma) by monitoring the *Recovery Elasticity* ( $\epsilon_{rec}$ ).

$$\epsilon_{rec} = \frac{d^2(\text{Recovery})}{d(\text{Stress})^2} \quad (3)$$

The Governance Response Function  $\mathcal{H}(\sigma)$  is:

$$\mathcal{H}(\sigma) = \begin{cases} \text{Allow Stress} & \text{if } \epsilon_{rec} > \text{Threshold}_{elastic} \quad (\text{Growth}) \\ \text{Throttle Stress} & \text{if } \epsilon_{rec} \leq \text{Threshold}_{elastic} \quad (\text{Trauma}) \end{cases} \quad (4)$$

*Implication:* The system allows stress only as long as the node demonstrates the capacity to recover stronger. If recovery efficiency drops, the system intervenes to prevent permanent deformation (plastic yield).

**Definition 8: Cumulative Toxic Load (The Boiling Frog Integral)**

To detect low-intensity, high-duration entropy (e.g., chronic neglect or abuse) that does not trigger acute alarms, the system calculates the accumulated entropic debt over time  $t$ .

$$L_{toxic}(t) = \int_{t_0}^t (\sigma_{cortisol}(\tau) + \sigma_{adrenaline}(\tau) - R_{recovery}(\tau)) d\tau \quad (5)$$

**Trigger Condition:**

$$\text{If } L_{toxic} > \text{Limit}_{bio} \implies \text{Initiate Audit (Override Privacy)} \quad (6)$$

**Definition 9: The Localized Computation Axiom (Edge Efficiency)**

To adhere to the Landauer Limit, data processing occurs at the Node level. The State is an Exception Handler, not a Monitor.

$$\text{Transmission}_{state} = \begin{cases} 0 & \text{if Local Entropy} < \text{Threshold} \\ 1 & \text{if Local Entropy} > \text{Threshold} \end{cases} \quad (7)$$